

Multiagent Adversarial Team Games

- Environment with multiple teams of agents, each with a shared team goal
- Outcome** at game's end describes which teams won
- Can be formalized as a **Markov Decision Process (MDP)** for each agent
- Has both cooperative and adversarial interactions



Example: RoboCup robotic football

Research Goals

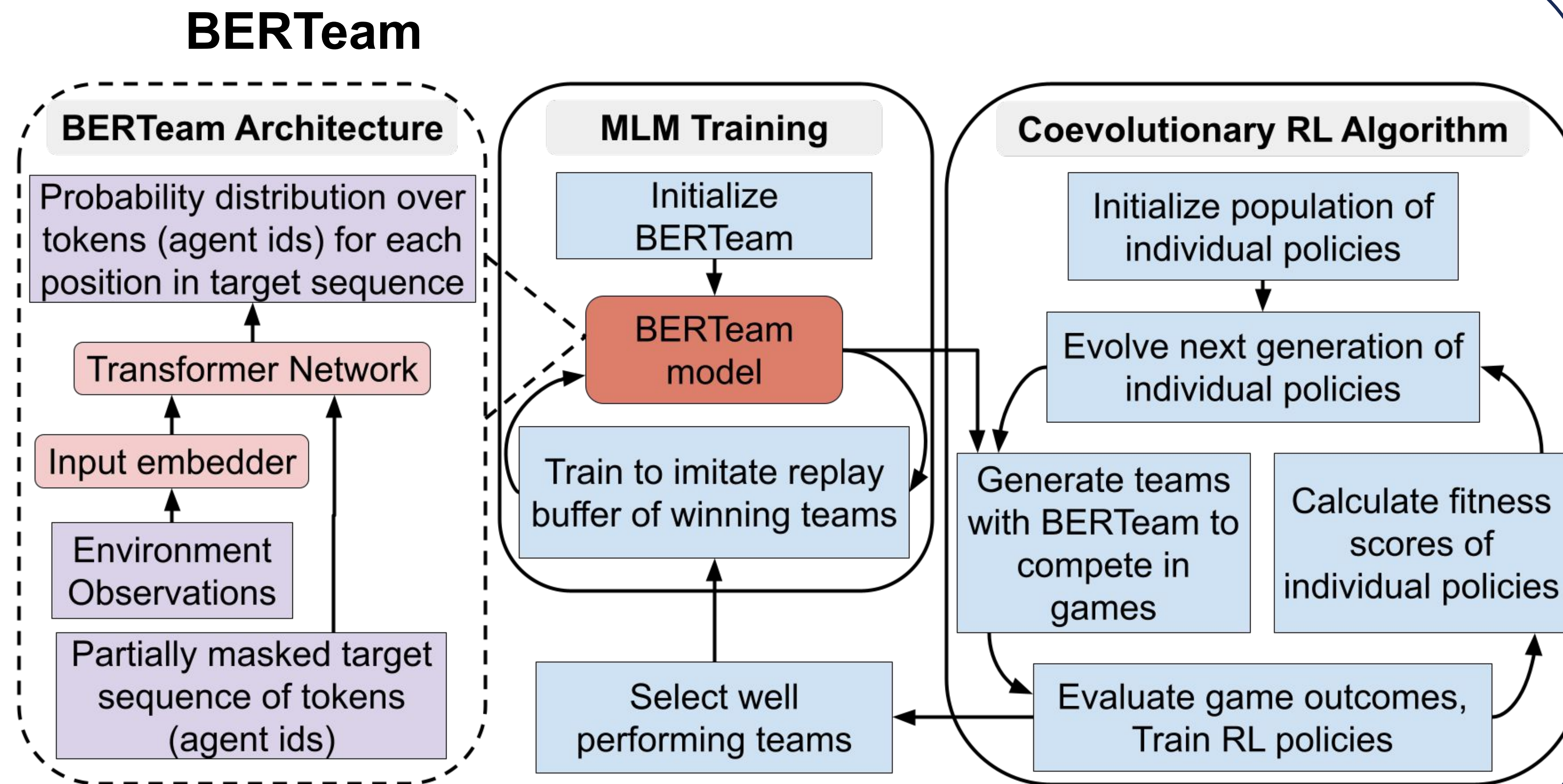
- Train individual agent policies in multiagent adversarial team games
- Select optimal teams from a set of trained agents
 - Team members must cooperate
 - Teams must perform well against various opponents
 - Difficult task, as for n agents there are $\Theta(n^k)$ sized k teams

Architecture/Training

- Selecting a team from a set of fixed-policy agents is equivalent to generating an 'agent sequence'
- We approach this with **BERTeam**, a transformer model trained through **Masked Language Modeling (MLM)**
 - Predict masked tokens in a sequence from context
- BERTeam is trained to imitate dataset of winning teams

Training alongside Coevolution

- Update individual policies alongside training BERTeam (instead of using fixed-policy agents)
 - BERTeam obtains stronger agents to select from
 - Agents gain experience being on stronger teams/playing against stronger opponents
- Coevolutionary Reinforcement Learning** improves a population of agents through **Reinforcement Learning (RL)**, and replaces agents that perform poorly



BERTeam architecture and training scheme

Experiments

We evaluate the performance of BERTeam experimentally in 2v2 games of **Pyquaticus**

Fixed Policies

- Trained BERTeam to generate teams from 7 fixed-policy agents
 - Offensive/Defensive agents of varying skill level
 - Randomly acting agent
- Calculated the team **Elo** of all possible teams to compare with BERTeam's distribution

BERTeam + Coevolution

- Trained BERTeam alongside Coevolutionary RL algorithm
 - Population: 50 **PPO** agents
- Estimated performance against fixed-policy agents
- Characterized trained policies as aggressive/defensive for analysis

Comparison with MCAA

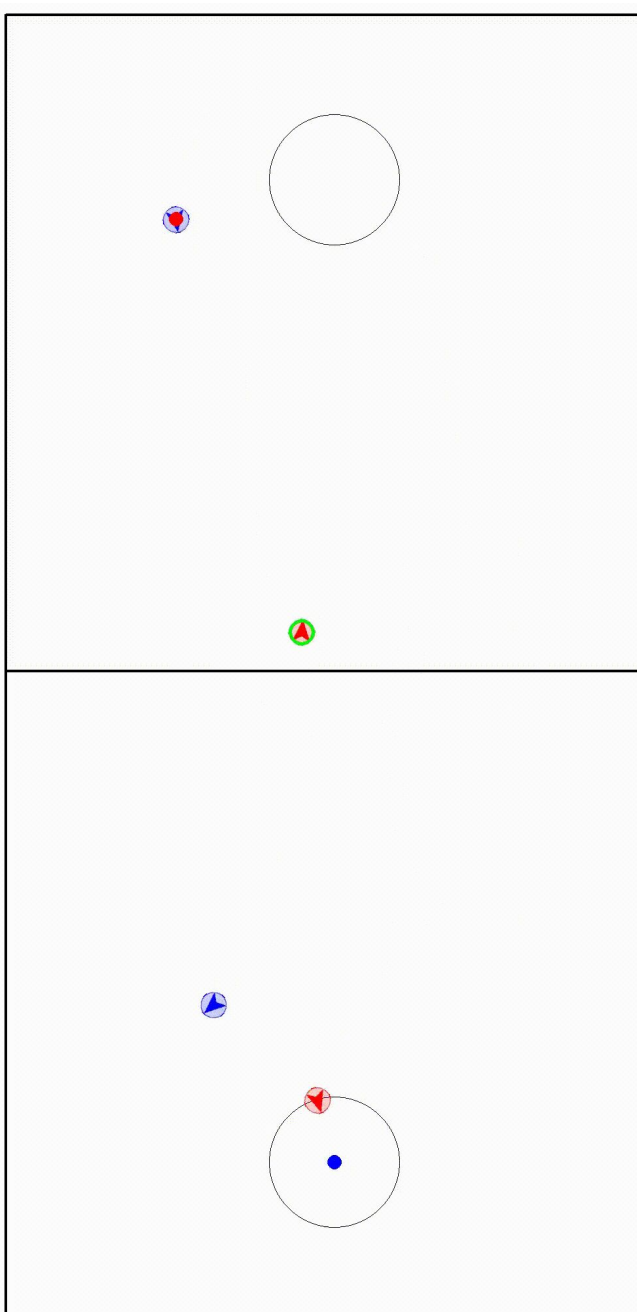
- Multiagent Coevolution for Asymmetric Agents (MCAA)**
- MCAA also decouples policy optimization (**MAP-Elites**) from its team selection, so we compare 4 hybrid algorithms
- Trained hybrid algorithms, evaluated games between teams selected from each

Pyquaticus

- Simulation of **Aquaticus**, a robot Marine Capture-the-Flag game
- k -v- k game, each team must capture a flag from opponent's base without getting tagged



Aquaticus:
Robotic Marine Capture the Flag



Pyquaticus
Screenshot

Results

Fixed Policies

- BERTeam chose the correct best team
 - {2,5}: One aggressive, one defensive agent
- BERTeam's top 7 similar to true ranking from Elos

Team	True Rank/Elo	BERTeam Rank/Occurrence
{2, 5}	1 1388	1 0.14
{2, 2}	2 1337	2 0.13
{2, 3}	3 1135	7 0.06
{1, 2}	4 1112	4 0.10
{0, 2}	5 1097	3 0.10
{2, 4}	6 1087	5 0.10
{2, 6}	7 1035	6 0.07
{0, 5}	8 975	13 0.03

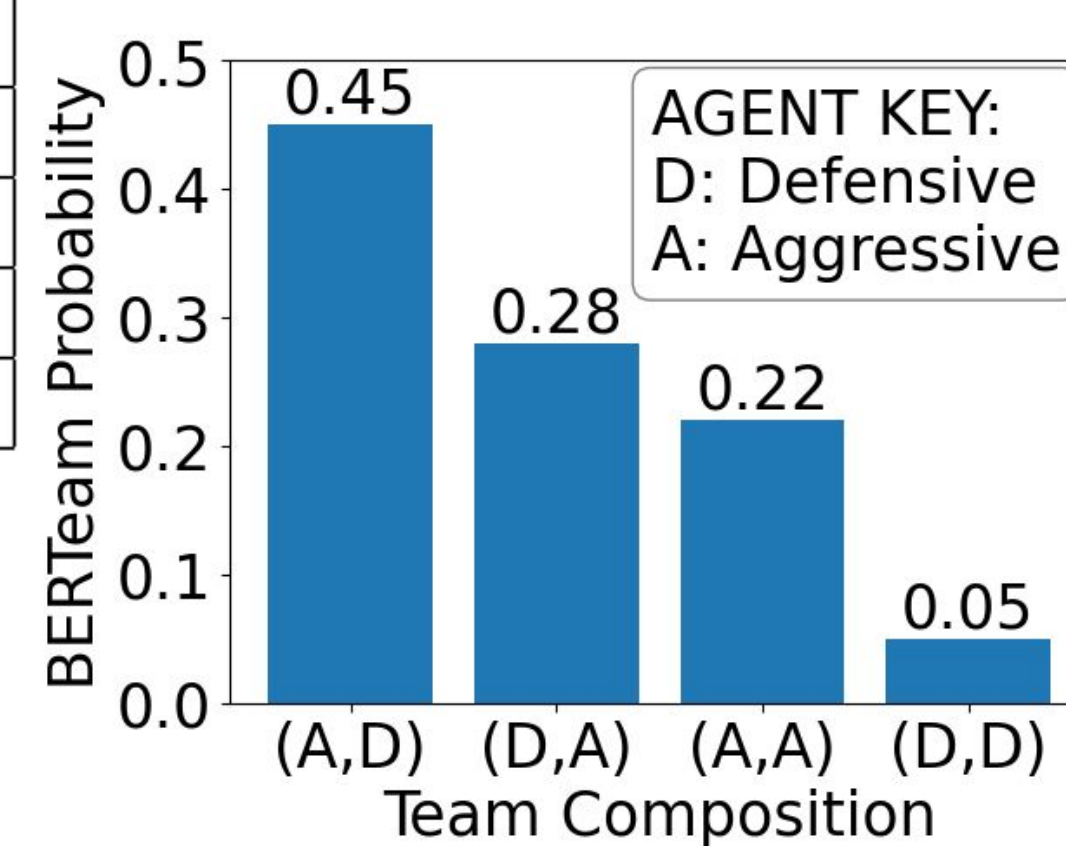
AGENT KEY:

Aggressive: Easy - 0, Medium - 1, Hard - 2
Defensive: Easy - 3, Medium - 4, Hard - 5
Random - 6

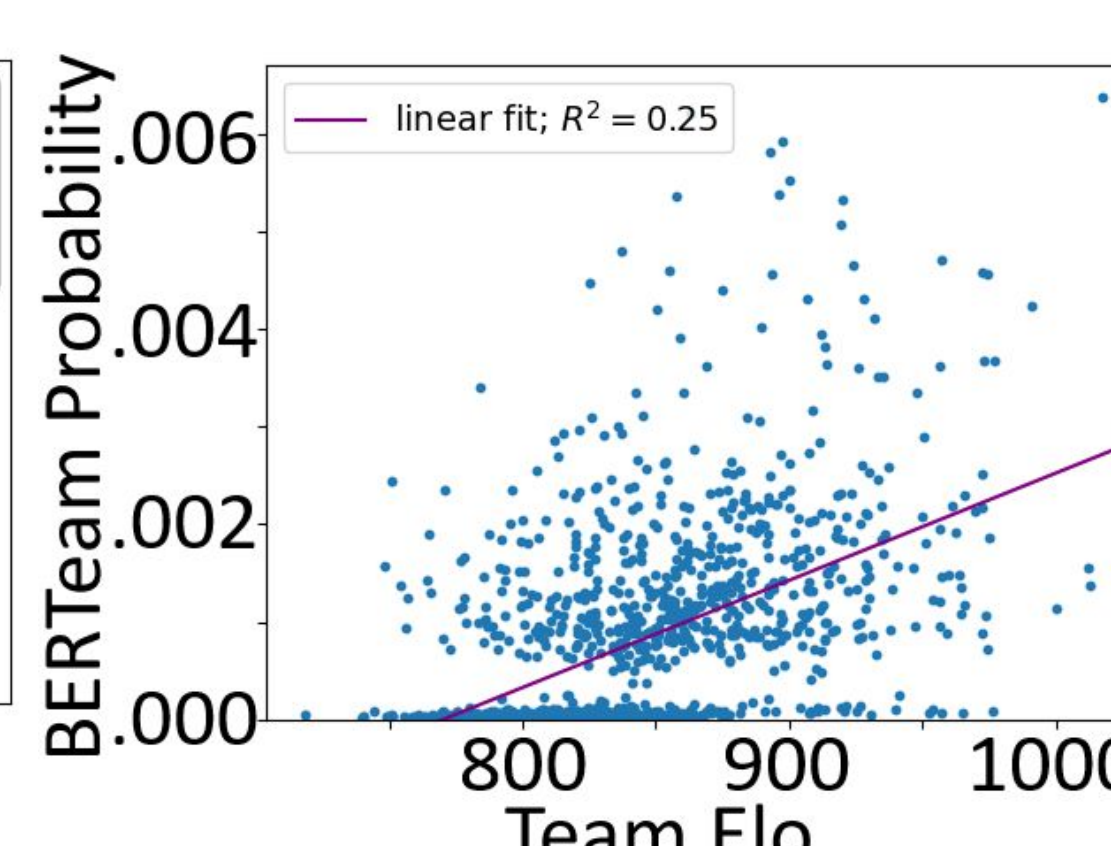
Comparison of true and predicted ranks

BERTeam + Coevolution

- Learned team composition is similar to best team with fixed-policy agent
 - One aggressive agent, one defensive agent
- BERTeam output correlates with team performance ($R^2 \approx 0.25$)
- Strongest team (≈ 1017 Elo) outperforms all fixed-policy teams that don't contain agent 2



BERTeam learned distribution on trained agents



Comparison with MCAA

- BERTeam outperforms MCAA independent of policy optimizer used
 - Compare algorithm Elo ($A[Elo]$) based on expected performance of a team selected by each algorithm
- BERTeam's network update takes significantly more time than MCAA's update

Policy Optimizer	Team Selection	$A[Elo]$	Avg. update time of	
			Agents	Team Dist.
Coevolution	BERTeam	919	13 s/epoch	46 s/update
Coevolution	MCAA	817	13 s/epoch	≈ 0 s/update
MAP-Elites	BERTeam	883	36 s/epoch	45 s/update
MAP-Elites	MCAA	809	35 s/epoch	≈ 0 s/update

Relative performance of hybrid algorithms

Conclusions

- BERTeam is able to optimize a distribution of teams from fixed-policy agents
- BERTeam trained with coevolutionary RL is competitive against strong opponents not in training data
 - This algorithm outperforms MCAA in the setting of Pyquaticus

Future Work

- Team selection may be phrased as a **normal form game** between *coaches*
 - Update BERTeam algorithm to approach a **Nash Equilibrium** in this game
- Generate teams given partial information about environment/opponents
- Test on games with larger team sizes
 - Transformers can comfortably generate size ≈ 512 sequences